

Writing Advice

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1 The Case for L^AT_EX

Before getting into more specific writing tips, it is worth explaining my own preference for L^AT_EX as a tool for scientific writing. It is, admittedly, complex at first, but the learning curve is well worth the value it provides in the long run. A common criticism of L^AT_EX is that it is difficult to collaborate on and to track changes within when you have multiple co-authors. This is one of the main reasons Microsoft Word has remained the preferred choice for many, largely thanks to its tracked changes feature, and it is precisely why I used Word throughout my own PhD. However, this convenience comes at a cost, since heavily edited Word documents are well known for slowing down, and even crashing, as they grow in size and complexity. I still shiver when I think back to my own PhD thesis that would crash every time I inserted a new figure. It became a reflex reaction to quickly hit save every time because I knew the freeze and crash was coming.

I first used L^AT_EX during my Master's thesis and, admittedly, had great fun doing so. However, I also found myself easily distracted by different formatting options and font types, rather than focusing on the actual content, which is exactly what L^AT_EX is supposed to help you avoid. It is only relatively recently, however, that I made the switch back to L^AT_EX with my own research group, so that we are all now working from the same template and are on the same page. To help with this, I created a simple L^AT_EX class file¹ that anyone in the group can use. The writer simply loads the class and focuses on the content, while all formatting decisions are handled centrally within the class file.

Overleaf was a major improvement in the usability of L^AT_EX, since it allowed for online collaboration in much the same way as Google Docs, directly addressing the earlier criticism about collaboration and tracked changes. It is free up to a certain point, but I have found it well worth the investment in the paid team version.² I won't go into all of its benefits here, but for article writing, collaborative review, and turning a PhD's worth of work into a full thesis, it makes the whole process considerably simpler. Modern AI tools also mean that the days of scouring Stack Overflow for a fix to a stubborn table or figure placement issue are largely behind us.

Overleaf's GitHub integration also allows you to properly version your documents, and to source figures directly from your working directory, so there is never any doubt about which version of a figure was actually inserted into the manuscript. Journals can sometimes prefer L^AT_EX submissions, and typesetting mistakes are greatly reduced as a result. Equations, in particular, look excellent and are straightforward to write once the syntax becomes familiar.

Another practical feature worth mentioning is the ability to comment out text using `%` or a `comment` environment, without actually deleting it. This is useful for supplementary material,

¹Available here: <https://www.overleaf.com/read/qsfbqcvnpxp#3487db>

²I am not sponsored by Overleaf, by the way.

such as a more in-depth discussion of a plot, that may not be needed for a concise paper submission but could be a nice addition if the same content is later expanded into a longer report or thesis chapter. Rather than deleting this material and having to rewrite it later, you can simply comment it out and leave it in place for whenever it is needed.

None of this is to say that L^AT_EX is the only tool worth using. I remain a fan of Microsoft Word for certain situations, particularly where detailed, line-by-line tracked changes are the priority, and an even bigger fan of Google Docs for quick, collaborative writing or planning, where the final formatting simply isn't important. Each tool has its place, and L^AT_EX is where I land for the polished, long-form scientific writing that is the focus of this document.

Hence, these are some personal notes on why I would recommend it.

2 Structuring a Document

I won't get into the specifics of designing your overall communication strategy here; [Baker \[2026\]](#) has already done an excellent job of breaking down how to “design” a piece of scientific communication into five stages. However, one thing I find particularly useful, once you have settled on your overall design and document structure is to add a further level of detail and structure within each individual section.

For example, in your introduction, you may decide that you want to touch on three key topics to “set the scene” for your work, and thereby properly introduce it. However, when you are faced with a blank page and a section simply titled “Introduction”, it is easy to panic at how empty it looks and just start writing for the sake of writing, purely to fill the space. After writing some text, you look back, perhaps with your advisor or co-authors, and realise that it is a little unstructured and rambles a lot. This is an easy trap to fall into.

What is far more useful is to first write a series of quick statements on what you want to say and achieve within a section, discussing these with your advisor or co-author before writing a single sentence. This gives you a clear breakdown of what the section is meant to achieve and say, and lets you draft much more targeted text from that point onward, greatly reducing the risk of going off on a tangent. It also helps you avoid spending time drafting text just for your co-authors to disagree, possibly delete it and suggest something else.

L^AT_EX Tip: Outlining with Comments

L^AT_EX is particularly well suited to this kind of outlining, since you can leave comments directly in the source text (using `%` or a `comment` environment) that never appear in the final PDF. I find this feature extremely useful when reviewing a co-author's or student's draft: a quick comment left above a paragraph telling me what they are trying to say helps me understand what they are trying to achieve, and how to guide them towards it. This is much harder when faced with a wall of text, after reading which you think: that was all technically correct, but I have no idea what they were trying to tell me. A well-placed comment avoids this entirely.

A further piece of advice is to write the conclusions first when trying to decide on the value of a piece of work. Doing so helps you identify the actual contributions, and avoid simply conveying “I did a lot of work, and here I am committing information to pages to make this abundantly clear.” When discussing these draft conclusions with your advisor or co-authors, it can help to revise these take-away points by asking “what am I trying to say?” or “who cares?”. These may sound like blunt questions, but they are useful for getting to the point. Only once you have a

clearer answer to these should you begin to properly design your structure and communication strategy.

3 Tenses

When writing a scientific article, the tense in which the text is presented is important. It is easy to get mixed up between past and present tense, especially if you write different parts of the article at different times. This can influence your train of thought, and you may find yourself describing methods in the present tense (e.g., A test specimen *is* created and subjected to several cycles of a loading protocol). However, after taking a break and returning to your writing, you might start thinking of the experiment differently, like something you completed in the past and begin using the past tense instead (e.g., The specimen *behaved* very well and *showed* good ductile behaviour). The result is a piece of text that may describe the activities clearly but contains inconsistent tenses, which can be confusing and is grammatically incorrect.

To avoid this, there is a simple set of rules³ that can be followed:

1. Actions or observations that were carried out in the past should be written in past tense. For example:
 - The test specimen *was* constructed and placed in the loading rig.
 - A standard loading protocol *was* applied and the specimen *exhibited* ductile behaviour.
 - A numerical model *was* constructed using OpenSees.
 - A survey *was* carried out.
2. Anything that is presented in the article itself should be written in the present tense. For example:
 - A numerical model was constructed using OpenSees and *is* shown in Figure 1.
 - A survey was carried out and the results *are* listed in Table 2.
3. Anything that is generally true in the broader context of the paper should be written in the present tense. For example:
 - Steel moment frames *are* a common lateral force resisting system.
 - Incremental dynamic analysis *is* a popular analysis method for quantifying building response.
4. Any mentions of literature or past work is written in the past tense. For example:
 - Smith (2000) *found* that steel moment frames performed well.

4 Active versus passive voice

Another common theme in technical papers today is the choice between using the active voice (e.g., We built a model) or the passive voice (e.g., A model was built). While both are valid

³I don't take credit for these. This was an issue that I had in my own writing as a student, and Prof. André Filiatrault kindly provided this set of rules to follow, which I have used ever since.

ways of conveying the same information, the passive voice has traditionally been considered more appropriate for technical writing. The passive voice gives an impersonal account of the work, and places the material at the centre, whereas the active voice places more attention on the writer or the person conducting the research. In recent years, the active voice has gained popularity in some research communities due to its simplicity and less formal tone. One valid criticism of the passive voice is that it can be easily abused to create vagueness, obscure details, or give the impression of having stated something without really saying anything at all.

Ultimately, it comes down to personal preference. Many grammar-checking tools will encourage the use of the active voice, but whichever style you choose, it's important to remain consistent. Personally, I prefer to use the passive voice, noting the irony that I wrote this text in the active voice.

5 Acronyms, abbreviations and symbols

5.1 Acronyms and abbreviations

In general, acronyms and abbreviations can be very useful for helping readers follow the text and avoiding overly repetitive phrasing. For example, the term reinforced concrete is almost synonymous with the acronym RC in many structural engineering papers. That said, defining too many acronyms can become overwhelming and may create confusion due to the number of terms the reader must keep track of.

As for when and where to define an acronym, a simple rule is: define it at its first mention. So, the first time you write reinforced concrete, you should introduce it as *reinforced concrete* (RC). From that point onward, you can use RC consistently. Needless to say, there's no point in defining an acronym if you don't plan to use it regularly. If the term appears only once or twice more in the document, it may not warrant an acronym, unless it refers to a well-known method, organisation, or concept (e.g., American Society of Civil Engineers (ASCE)).

In some cases, especially in theses or dissertations, authors include a complete list of symbols and acronyms at the beginning of the document. While this is certainly useful, it's not always practical or required. Therefore, the advice here applies to documents where such a list has not been implemented.

L^AT_EX Tip: Managing Acronyms with the `acro` Package

Rather than manually tracking where each acronym has already been defined, consider using the `acro` package to handle this for you. Acronyms are defined once in a separate file, and from then on can be referred to anywhere in the text using:

- `\ac{IM}` for singular mentions, e.g. intensity measure (IM)
- `\acp{IM}` for plural mentions, e.g. intensity measures (IMs)

This keeps all acronyms centrally defined and makes it much easier to keep track of your full acronym list.

Another common misconception is capitalising the words when defining an acronym, such as writing *Reinforced Concrete* (RC) instead of *reinforced concrete* (RC). This is unnecessary and should generally be avoided unless a proper noun is involved. Otherwise, it is an indicator that your text is AI-generated.

A further point concerns plural forms. If you define an acronym in the singular form (e.g., moment-resisting frame (MRF)) and later refer to it in the plural (e.g., moment-resisting frames (MRFs)) there is no need to redefine it. Simply using MRFs is sufficient and understood.

An exception to these conventions applies to figure captions and axis labels. Excessive use of acronyms in figures can reduce clarity. Since the goal of a figure is to be as self-explanatory as possible, it is often better to write out the term fully (e.g., Engineering demand parameter (EDP)) on axis labels or in figure captions. Of course, this must be balanced with avoiding clutter or overly long labels, so some ad hoc judgment is sometimes necessary.

Finally, a key consideration is where acronyms should be defined. A well-structured technical article typically includes three main parts: the abstract, the main body, and the summary and conclusions. Each of these sections should be as self-contained as possible, allowing readers to understand the content without needing to refer elsewhere. Therefore, if an acronym is used in any of these sections, it should be defined there, even if it has already been defined elsewhere in the document. For example, if an acronym appears in the abstract, it should be redefined the first time it appears in the main body, and again in the conclusions if used there. While this may seem redundant, it improves accessibility for casual readers who may only browse the abstract or conclusions without reading the full article.

5.2 Symbols

On the issue of symbols, these should be introduced and explained fully in the text by saying something like: Newton’s second law states that the force, F , is given by:

$$F = ma \tag{1}$$

where m is the mass of the object and a is the acceleration. Defining the symbol F above should come directly after mentioning it (i.e., the force, F , or the mass, m), where I tend to separate these with commas. Others use parentheses (i.e., the force (F) or the mass (m)), which can also be fine but again, a personal preference. Symbols should also always be written in math typeface and not regular typeface (i.e., F and not F).

L^AT_EX Tip: Using `acro` for Symbols

Another tip for L^AT_EX users is that the `acro` package can also be “abused” to define symbols, not just acronyms. For example, $Sa(T)$ can be defined under the key `SaT` and then simply referred to as `\ac{SaT}` at every mention. This has the advantage that the symbol is properly defined at its first mention, while remaining easy to call upon at every subsequent mention in the manuscript.

6 References

Citing relevant literature is a fundamental part of scientific writing. It gives credit to prior work, positions your contribution within the broader field, and gives readers a trail to follow if they want to explore the topic further. Below are some practical points on how to handle in-text citations and how to reference illustrative examples.

Before getting into the mechanics, it is worth clarifying when a citation is actually needed:

- A statement of general, widely accepted knowledge does not require one, for example: *Reinforced concrete is a composite material that combines the compressive strength of concrete with the tensile strength of steel.*

- A statement that reports a specific finding, method, or claim attributable to a particular study does require one, for example: *Steel moment-resisting frames have been shown to sustain significant residual drifts even under moderate seismic demands (Smith, 2010).*
- The same claim can also be attributed directly to its authors using a narrative citation, for example: *Smith (2010) found that steel moment-resisting frames sustained significant residual drifts even under moderate seismic demands.*

As a simple test, if a claim could reasonably be challenged or asked “says who?”, it needs a citation; if it would be common knowledge to anyone in the field, it does not.

L^AT_EX Tip: Citations with natbib

Using the `natbib` package, together with `\bibliographystyle{plainnat}` and `\bibliography{references}`, in-text citations can be produced as follows:

- `\citet{AuthorYear}` to get: Author (Year)
- `\citep{AuthorYear}` to get: (Author, Year)
- `\citep[e.g.,][]{AuthorYear}` to get: (e.g., Author, Year)

Another point worth discussing is how to cite example references in support of a general statement, without implying that the works cited are the only ones to have addressed the topic. This is typically done by prefacing the citation with “e.g.” (i.e., `\citep[e.g.,][]{AuthorYear}`), which serves a few purposes at once. First, it gives credit to the fact that the topic has already been discussed by others, avoiding the impression that you are the one proposing the notion. At the same time, citing only a handful of subjectively chosen key references, rather than attempting to be exhaustive, keeps the citation list focused and manageable. Prefacing these with “e.g.” signals to the reader, and to reviewers, that the list is illustrative rather than complete, so nobody has grounds to be disgruntled that another relevant paper, possibly even their own, was not included. It also protects you a little from coercive citation requests where a reviewer of your paper asks you to expand your cited literature with some specific examples, usually containing a common co-author. Your response to such request can be “We cited some paper relevant to topic, just not those specific ones.” Ultimately, a paper should be informative and give credit where it is due, but it does not need to provide an exhaustive list of every piece of work on a topic, regardless of its actual relevance or value. Exceptions to this would be review papers whose purpose is to be exhaustive.

A related point concerns which reference to cite when an idea, method or result could reasonably be attributed to more than one source. Where possible, it is best practice to cite the original, or seminal, source rather than a later paper that simply repeats or builds on it. This gives credit to the work that actually originated the contribution and directs the reader to the most authoritative account of it, rather than a second-hand summary. That said, this is not always straightforward, since seminal papers can be difficult to track down, published in venues that are no longer easily accessible, or written in a different language. In such cases, it is reasonable to cite a well-regarded review or a more recent paper that gives a clear account of the original work, while still acknowledging the seminal source wherever possible.

7 Summary and conclusions

Another important aspect of academic writing is how to effectively summarise and conclude an article. In modern academic publishing, this section plays a crucial role, as many readers will not take the time to read the entire paper and thoroughly digest the findings and arguments presented in the main body on. More often, researchers will simply read the abstract, possibly the introduction, and the summary and conclusions. Based on this, they may decide that the topic addressed is of direct interest and use to them, so it is worth investing some time in reading the full article. Therefore, this final section should do precisely that: summarise the work carried out and clearly present the main findings. Think of it as almost a sales pitch for why your work was is a novel contribution and why the paper is worth investing a few hours to read in full.

A strategy I've used for many years, which is common among many researchers and not something unique to me, is to divide this section into two distinct parts. The first part consists of a single paragraph, no longer than half a page, which briefly summarises the work presented in the paper. This is all written in the past tense and directly states what was done in the paper, without elaborating further or offering concluding insight. This is then typically followed by the sentence: "*Based on this, the following conclusions can be drawn:*" and a concise set of bullet points. These bullet points should clearly state the key takeaways, which are what the reader should remember after going through 20 or so pages of detailed work and discussion.

In fact, when considering whether a piece of research is worth developing into a publication, it can be helpful to try writing this section first. If you find that the bullet points are few, vague, or simply restate the obvious, it's usually a sign that the paper may not add much value or generate significant interest. This doesn't necessarily mean it won't be accepted for publication, but it may indicate that it's not the kind of paper others will find especially useful.

Here is an example of this approach from [O'Reilly and Shahnazaryan \[2024\]](#) with the summary, followed by the bullet point conclusions:

This paper presented a comparative study of two methods to develop vulnerability functions for regional seismic risk modelling. These methods were based on analytical methods and involved several key assumptions to make them widely applicable and computationally highly efficient. The first of these methods was based on the fragility analysis of displacement-based demands on an equivalent single-degree-of-freedom (SDOF) model, which, in tandem with loss ratios, returns vulnerability functions. A second and alternative approach described here utilises an equivalent SDOF model to estimate both displacement- and acceleration-based demands at each building level. Combining these demands with storey loss functions (SLFs) leads to the estimation of vulnerability at each storey and, ultimately, the vulnerability function of the building. The two methods were described in detail, and a comparative case study example was presented to scrutinise their similarities and differences. Some final remarks on the wider implications and possible future directions were also presented.

Based on the work presented here, the following conclusions can be drawn:

- The SLF-based approach is a more detailed and robust methodology compared to the widely-used fragility and damage-to-loss ratio-based approach. The SLF-based approach is more detailed but does not carry the burden of excessive extra computation or effort. In fact, when simplified, the SLF-based approach can be reduced to give essentially the same result as the fragility-based approach.
- The SLF-based approach directly considers both peak storey drift (PSD)- and peak floor acceleration (PFA)-based demands at all storeys of the building. This breakdown allows for a more detailed consideration of non-structural elements and identifies which performance groups at which building location contribute most to the expected loss.
- The issue of collapse and demolition are handled more directly, which can be useful for decision-making in issues beyond economic losses, such as estimating casualties or identifying when retrofitting interventions may be futile.
- While SLFs are needed for different building taxonomies, the generalised format may be a prudent strategy given that a physics-based approach to modelling the building component fragility can be adopted, followed by a rational approach to estimating the expected repair costs.
- The uses of this approach extend beyond vulnerability modelling and can also aid insurance companies in deciphering and establishing a relative scale of expected damage and loss distribution within a building to determine more reasonable tariffs.

Overall, this extension of the existing to the proposed approach is without much additional computational cost and can represent an enhanced level of quality in the output decision variables used in regional seismic risk assessment.

8 Figures and graphics

When presenting results or data, plotting figures is almost always a very effective way to convey information. While there are many guides on how to create good versus poor plots (see Jack Baker's [YouTube video](#) or [Baker \[2026\]](#) with some excellent pointers on making effective figures), one aspect that is sometimes overlooked is the relative dimensions and sizing of figures.

This may seem trivial, as once you define and create a figure, you can simply scale and crop it in Microsoft Word to make it appear correctly in the final document. While this approach works in many cases, it can lead to problems when publishing through journals or dealing with typesetting staff. Many of us have experienced the situation where a manuscript is accepted, and we eagerly await the proofs, only to discover that the figures are awkwardly scaled, either

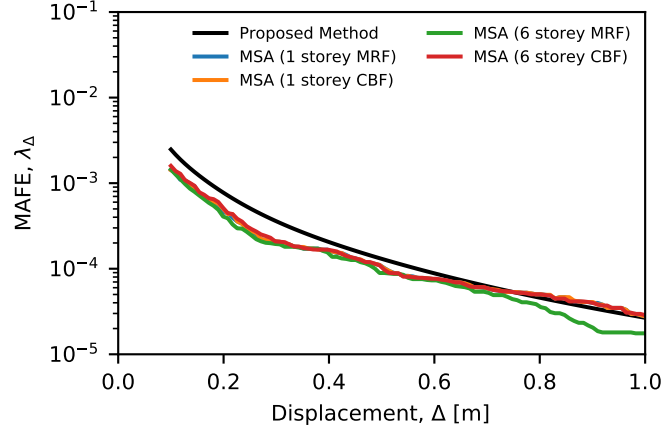


Figure 1: Example plot with no scaling

too large or too small, and not at all as originally submitted. What has likely happened is that the publisher’s typesetters imported your original as-is figure files without adjusting their size or formatting like you did to make it look nice in your final version. As a result, the figures may end up barely legible and visually unappealing.

To avoid this, it is helpful to be a bit more prepared and provide figures in vector format (e.g., .PDF, .SVG) that are sized appropriately for 100% scaling—rather than relying on arbitrary scaling (e.g., “eyeballed” adjustments in Microsoft Word). This ensures that line weights and font sizes appear clean and professional in the final version.

A good figure dimension for a single plot is around 6 cm tall by 8 to 10 cm wide, with a font size of approximately 8 pt to 10 pt maximum. Slightly smaller font sizes like 6 pt for axis ticks and legend text can help the axis labels stand out and reduce visual clutter. While there’s no fixed rule for these numbers, I’ve found them to work well in practice, but the dimensions do approximate the “Golden Ratio”⁴, which is somewhat reassuring.

Figure 1 is an example of such an approach from O’Reilly et al. [2022], where the plot was prepared in Python and the source code as given as shown below. Here it can be seen how the dimensions are initially set to about 9 cm by 6 cm (Python works in imperial units, hence 2.36 inches is around 6 cm, and 1.5 times 6 cm is 9 cm). The default font size is set to 8 pt, where the legend font is set to 6 pt. L^AT_EX syntax is used to display symbols in the axis labels and the figure is saved as a vector-based PDF.

```

1 plt.figure(figsize=(1.5 * 2.36, 2.36))
2 plt.rcParams.update({'font.size': 8})
3
4 # Plotting functions to display data
5
6 plt.xlabel(r'Displacement,  $\Delta$  [m]')
7 plt.ylabel(r'MAFE,  $\lambda_{\Delta}$ ')
8 plt.yscale('log')
9 plt.ylim([1e-5, 1e-1])
10 plt.xlim([0, 1])
11 plt.legend(ncol=2, fontsize=6, frameon=False)
12 plt.savefig('plots/Demand-Hazard-Curve-Case-Study.pdf', bbox_inches='
    tight')
```

⁴https://en.wikipedia.org/wiki/Golden_ratio

```
13 plt.close('all')
```

Although I no longer use Matlab and have since passed to Python, I was an avid user during my PhD days, and had several tools and shortcuts that helped make nice plots and data presentation. A GitHub repository is [available online](#) with much of what was described above presented there also.

In addition, when creating graphics, I often use [Inkscape](#), which is a free online software. One of the first steps I take is to enable a grid and place an A4 sheet behind the drawing area for reference. This helps guide the image creation relative to the actual size of an A4 page, so that the final output is neither too small nor too large and does not require any of the ad hoc adjustments previously mentioned.

References

- Jack W. Baker. *Communication by Design: An Engineer's Guide to Sharing Your Research*. Coast Range Press, 2026. ISBN 9798994527719.
- Gerard J O'Reilly and Davit Shahnazaryan. On the utility of story loss functions for regional seismic vulnerability modeling and risk assessment. *Earthquake Spectra*, 40(3):1933–1955, aug 2024. ISSN 8755-2930. doi: 10.1177/87552930241245940. URL <https://journals.sagepub.com/doi/10.1177/87552930241245940>.
- Gerard J. O'Reilly, Hiroshi Yasumoto, Yoshitaka Suzuki, Gian Michele Calvi, and Masayoshi Nakashima. Risk-based seismic design of base-isolated structures with single surface friction sliders. *Earthquake Engineering & Structural Dynamics*, 51(10):2378–2398, aug 2022. ISSN 0098-8847. doi: 10.1002/eqe.3668. URL <https://onlinelibrary.wiley.com/doi/10.1002/eqe.3668>.